Ocean Ecology Lab
Hatfield Marine Science Center
Oregon State University

To whom it may concern:

I am writing in response to the opening for the Postdoctoral research position in marine mammal population dynamics. I am currently a PhD candidate in the Institute for the Oceans and Fisheries at the University of British Columbia, and will be completing my degree by February 2027. I am particularly passionate about using statistical methods to model marine population dynamics with the goal of improving conservation and management. My academic experience has evolved to become increasingly quantitative, and throughout my PhD I have employed a diverse array of statistical methods, including multivariate, Bayesian, and hierarchical model structures, to support analysis of marine populations in space and time. Outside of my academic work, I have also worked with a number of federal agencies, both in the United States and Canada, in a variety of quantitative capacities.

Before my PhD, I was a Sea Grant Knauss Marine Policy Fellow, where I worked for the National Stock Assessment Program at NOAA Fisheries. In this position, I assisted in coordination, communication, and organization between fishery management regions to build a new stock assessment model framework, the Fisheries Integrated Modeling System (FIMS), which gave me insight into national collaborative efforts within NOAA Fisheries, and perspective about regional differences in fisheries science and management.

During my PhD, I have held a number of positions focused on research questions outside the scope of my dissertation work, which have expanded my quantitative skills and experience working with stakeholders. In the second year of my PhD, I participated in, and co-authored, an operational stock assessment for Shortspine Thornyhead (*Sebastolobus alascanus*) for the NOAA Northwest Fisheries Science Center. This process involved an age-structured population model, and a review by the Pacific Fishery Management Council, which both advanced my population modeling skills and exposed me to the operational science to management process in the U.S.

For much of my PhD I have been contracted as a research assistant on a project funded by the U.S. Army Corps of Engineers to assess survival of juvenile Chinook salmon (*Oncorhynchus tshawytscha*) over dams in Oregon. Results from my analyses have contributed directly to dam management, and the project allowed me to lead workshops and collaborate within a complex management system that involves competing objectives and diverse stakeholders. Similarly, I also worked with the Pacific Salmon Commission, a bilateral agency managing transboundary salmon stocks in the United States and Canada, to improve monitoring methods of sockeye (*Oncorhynchus nerka*) and pink salmon (*Oncorhynchus gorbuscha*) migrating upriver by using statistical models to compare data from different methods, and assess their ability to predict total abundance. In this position, I gained a more comprehensive understanding of the interconnections between management panels, technical working groups, and fishery stakeholders.

Of particular relevance to this position, I worked for the Department of Fisheries and Oceans Canada (DFO) where I assessed harbour seal (*Phoca vitulina*) diet sampling programs, modeled spatial temporal dynamics in seal diets, and compared diet estimates and dynamics with ecosystem model outputs. This work required use of spatial-temporal models, and expanded my knowledge of pinniped trophic and spatial ecology. In addition to this seal research, my PhD dissertation utilized population models for salmon and steelhead to explore drivers of long-term productivity declines. My findings indicated increased predation mortality by pinnipeds as a result of their dramatic population growth in B.C. As such, though most of my research to date has focused on fish population dynamics, my dissertation work has necessitated an understanding of pinniped ecology and population biology, and sparked my interest in further understanding the population and ecosystem-level consequences of marine mammal population recoveries. I believe my academic training in statistical ecology and fisheries biology, and my experience modeling fish population dynamics, provides a solid foundation for this Postdoctoral position, and would support an exciting expansion of my knowledge and skills.

Thank you in advance for your time and consideration,
Haley Oleynik

References

Dr. Murdoch McAllister, *PhD supervisor*	m.mcallister@oceans.ubc.ca
Dr. Carl Walters, *Academic Mentor*	c.walters@oceans.ubc.ca
Kristan Blackhart, former *NOAA supervisor*	kristan.blackhart@noaa.gov

Selected Publication

Oleynik, H.A., J.J. Bizzarro, E.A. Hale, A.B. Carlisle. 2024. Environmental drivers of biogeography and community structure in a Mid-Atlantic estuary. Oecologia. https://doi.org/10.1007/s00442-023-05500-z

Haley A. Oleynik

PhD Candidate
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EDUCATION

Colby College, *Waterville, Maine*, **B.A.** Biology & Sociology 2015

University of Delaware, *Lewes, Delaware*, **M.S.** Marine Science 2020

University of British Columbia, *Vancouver, BC*, **PhD** Fisheries Science in progress

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RESEARCH EXPERIENCE

PhD Candidate Sept 2021 - Present

Institute for the Oceans and Fisheries University of British Columbia

My PhD research involves population modeling of 27 stocks from five species of Fraser River salmon and steelhead (*Onchorynchus spp.*) to identify environmental, ecological, and anthropogenic drivers of productivity decline, with the goal of understanding regional, multi-species population patterns and incorperating ecosystem information into fisheries and ecosystem management.
Skills: Statistical modeling, Bayesian modeling, R programming, Git/Github, stakeholder engagement

Graduate Research Assistant Jan 2024 - Present

United States Army Corps of Engineers

As a part-time research assistant I conduct statistical modeling, using Bayesian mark-recapture models in JAGS, zero-inflated binomial regressions, and generalized additive models, of Chinook salmon (*Oncorhynchus tshawytscha*) passage success and survival through dams in the Willamette River system in Oregon.
Skills: Statistical modeling, Bayesian modeling, JAGS programming, data analysis, stakeholder engagement, R programming

Research Analyst Feb 2026 - May 2026

Pacific Salmon Commission Vancouver, B.C.

During a three month contract I analyzed salmon monitoring data to statistically evaluate the use of a new monitoring method for salmon passage through Hells Gate, a narrow gorge which poses passage challenges for salmon migrating up the Fraser River to their spawning grounds
Skills: Statistical modeling, R programming, Git/Github

Aquatic Biologist II Jun 2025 - March 2026

Department of Fisheries and Oceans Canada Vancouver, B.C.

As a contract research biologist, I used used empirical harbour seal diet data and modeled diets from Ecopath with Ecospace ecosystem models to statistically validate model outputs and inform field sampling programs
Skills: Spatio-temporal modeling, ecosystem modeling, EwE, TMB programming, R programming

M.S. Student Sept 2018 - July 2020

School of Marine Science and Policy University of Delaware

My Master's research involved analyzing data from a state-run trawl survey using multivariate statistical methods to describe the fish and invertebrate community in Delaware Bay, identify physical drivers of community composition, and evaluate long-term changes in the community
Skills: Fishery independent data analysis, multivariate statistics, timeseries analysis, R programming

Undergraduate Senior Honors Thesis Sept 2014 - May 2015

Department of Sociology Colby College

Conducted in-depth interviews of harvesters to understand social and community implications of the changing American lobster (*Homarus americanus*) population, and subsequent management, on the lobster fishing industry in Maine

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PROFESSIONAL EXPERIENCE

Teaching Assistant Sept 2023 - Present

Master of Data Science Program University of British Columbia

For four years, I have been a part-time teaching assistant for the following MDS statistics and computation courses:

DSCI 523: Programming for Data Manipulation
DSCI 552: Statistical Inference and Computation I
DSCI 561: Regression I
DSCI 562: Regression II
DSCI 553: Statistical Inference and Computation II
DSCI 554: Experimentation and Causal Inference

Teaching Assistant Spring 2026

Ecosystem Modeling with EwE University of British Columbia

I served as a teaching assistant for a graduate-level Ecosystem modeling course (FISH 501), which taught the use of Ecopath with Ecosim for marine ecosystem modeling.

Graduate Writing Consultant Sept 2022 - April 2024

Center for Writing and Scholarly Communication University of British Columbia

Provided written and in-person consultations to graduate and undergraduate students on written work including class assignments, manuscripts, and dissertations

NOAA Sea Grant John A. Knauss Marine Policy Fellow Feb 2021 - Feb 2022

National Stock Assessment Program NOAA Fisheries

As a Knauss Fellow I supported the National Stock Assessment Program by organizing a technical community, developing the Fisheries Integrated Toolbox, assisting in coordination of a new national stock assessment modeling framework and facilitation regional communication, to advance the NOAA stock assessment enterprise

Communications Committee Chair Oct 2022 - June 2026

Committee Member June 2015 - Oct 2022

Association of the Marine Laboratories of the Caribbean

In this volunteer position I managed internal and external communication for the AMLC, a nonprofit which connects marine laboratories, scientists, students, and stakeholders in the Wider Caribbean Region

North Pacific Groundfish Observer June 2016 – Jan 2018

Saltwater Inc. / NOAA Fisheries

As a fisheries observer I collected data on catch, bycatch, and fishing effort on board commercial Alaskan fishing vessels to be used by the National Marine Fisheries Service to monitor fish populations, set catch limits, and manage the commercial Alaskan fisheries

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OUTREACH AND ENGAGEMENT

REDI Committee **Student rep**, UBC Institute for the Oceans and Fisheries 2022-present

Society for Women in Marine Science **Co-Chair**, University of Delaware Chapter 2019-2020

Graduate Student Association **Co-Chair**, UD School of Marine Science and Policy 2019-2020

Hutton Junior Fisheries Biology Internship **Reviewer**, American Fisheries Society 2024

John A. Knauss Marine Policy Fellowship **Reviewer**, NOAA Sea Grant 2022, 2024

PUBLICATIONS

Brodziak, J., M. Supernaw, B. Li, C. Stawitz, **H.A. Oleynik**, T. A'mar. 2025. Engineering a prototype for a next generation stock assessment modeling platform. Fishery Bulletin. 123(2). 88-104.

Oleynik, H.A., J.J Bizzarro, E.A. Hale, A.B. Carlisle. 2024. Environmental drivers of biogeography and community structure in a Mid-Atlantic estuary. Oecologia.

Zahner. J.A, M.A. Heller-Shipley, **H.A. Oleynik**, S.G. Beyer, P-Y. Hernvann, M. Véron, A.N. Odell, J.Y. Sullivan, A.L. Hayes, V.G. Gertseva, K.L. Oken, O.S. Hamel, M.A. Haltuch. 2023. *Status of Shortspine Thornyhead (Sebastolobus alascanus) along the US West coast in 2023*. Pacific Fishery Management Council, Portland, Oregon. 139 p.

Blackhart, K., and **H.A. Oleynik**. 2023. *Setting targets and analyzing data gaps for U.S. fish assessments*. NOAA Tech. Memo. NMFS-F/SPO-242, p. 77

Oleynik, H.A. 2022. *Living Resources: Fish Population Trends in Delware Bay: Climate and Community*. Technical Report for the Delaware Estuary and Basin, J. Shinn, L. Haaf, L. Morgan, and D. Kreeger (eds). Partnership for the Delaware Estuary.

ICES (2022): Working Group on Technology Integration for Fishery-Dependent Data (WGTIFD; outputs from 2021 meeting). ICES Scientific Reports. Report.

Mathews, J.A., E.A. Hale, **H.A. Oleynik**, et al. (2022) *Exploring Trends in Abundance of Young-of-the-Year and Age-1 Atlantic Croaker, Black Drum, Spot, and Weakfish in Relation to Salinity, Temperature, and Large-Scale Climatic Signals in a Mid-Atlantic Estuary.* Transactions of the American Fisheries Society, 151(2), 150–171.

AWARDS AND FUNDING

Partnership Engage Grant *Social Sciences and Humanities Research Council, Canada* 2024

Buck Suzuki Legacy Bursary Award *T. Buck Suzuki Foundation* 2022

Frances Severance Thesis Distinction Award *University of Delaware* 2021

Daiber Fellowship *University of Delaware* 2019

Women of Promise Award *University of Delaware* 2019

Okie Fellowship *University of Delaware* 2018

PRESENTATIONS

"Impacts of Dam Operations on Mortality and Injury of Juvenile Chinook Salmon at Cougar Dam" Willamette Science Review 2026

"Population-level survival impacts of Harbour seal foraging on juvenile Pacific salmon, and the mechanistic links" PICES 2025

"Drivers of Productivity in Fraser River Salmon" American Fisheries Society WA-BC Chapter Meeting 2025

"Incorperating Human Dimensions into EBFM: Understanding Socioeconomic Consequences of Ecological Change" American Fisheries Society Annual Conference 2024

"Integrating Electronic Monitoring Data into U.S. Fish Stock Assessments" World Fisheries Congress 2024

"Fisheries Science and Management" Guest Lecture, George Washington University Marine Ecology Course (BISC 3454) 2024

"Long-Term Shifts in the Fish and Macroinvertebrate Community in a Mid-Atlantic Estuary" American Fisheries Society Annual Conference 2021

"Changes in a Mid Atlantic Estuary: Long Term Shifts in the Fish and Macroinvertebrate Community in Delaware Bay" NOAA Central Library 2021

"Long-Term Shifts in the Fish and Macroinvertebrate Community in a North American Estuary" World Fisheries Congress 2021

"Changes in a Mid-Atlantic Estuary: Long-Term Trends in Environmental Conditions and Marine Community in Delaware Bay" American Fisheries Society Mid-Atlantic Chapter Meeting 2020

EXPERTISE

Computing

Advanced in R, WinBUGS, JAGS, Git/GitHub; experience with C++, Stan, Stock Synthesis, Template Model Builder (TMB), VAST

Field & Lab

AAUS Scientific Diver, SDI Rescue Diver, small boat operation, random field sampling, otolith collection, shark tagging, marine animal husbandry